Smart Energy-Saving Room Lighting System

Course: CPE 301: Embedded Systems Design

Project Type: Individual, Aansh Sachdeva

1. Project Introduction

In many residential and commercial buildings, it has been observed that lights are often left on in empty rooms, resulting in a waste of energy and increased energy bills. The aim of this research is to develop an automatic Smart Energy-Saving Room Lighting System. The system works by using an embedded system which detects the presence of people in the room as well as the presence of light. Hence, the system doesn't involve any human input to switch the lights on and off, as long as there are people in the room and the light available is not adequate.

In this system, the use of a state machine that guarantees efficient system operation, gives feedback to the user via the LCD screen, and identifies faulty sensors with the safety error recovery mode is essential. The lighting of buildings consumes a substantial amount of energy. In the smallest possible ways, even just forgetting to turn off lights in empty rooms results in energy wastage. By not depending on humans to manually turn off the lights, the energy used will be minimized.

2. Proposed System

The system makes use of an ATMega 2560 microcontroller to control the lighting of the room which is represented by an LED. The system uses two sensors to sense the surrounding environment; one being the Ultrasonic Sensor that senses the presence of humans in the room, and the other being the Photoresistor that senses the amount of ambient light.

Core Hardware Components

- Microcontroller: ATMega 2560
- Sensors: HC-SR04 Ultrasonic Sensor and Photoresistor (LDR)
- Actuator: High brightness LED (Room Light)
- User Interface: 16x2 LCD Display, 4 State LEDs(Red, Yellow, Green Blue), Push Buttons (ON, OFF, RESET)
- Logging: RTC DS1307 Module

2.1 System Schematic Diagram

The following table illustrates the interaction between the microcontroller and peripheral hardware:

Components	Purpose	Justification
ATMega 2560	Controller	Multiple GPIOs + external interrupt support
Ultrasonic Sensor	Occupancy Direction	Distance measurement for occupancy
LDR	Light Sensing	Analog light level reading via ADC
LCD Display	User Interface	Real-time state and sensor feedback
RTC DS1307	Event Logging	Precise timestamps for state transitions
State LEDs (x4)	State Indicators	Visual indication of current FSM state
Push Button (x3)	User Input	Manual ON, OFF, and RESET control

Pin Connection

Components	Pin	Register
HC-SR04 TRIG	Pin 9	PH6
HC-SR04 ECHO	Pin 10	PB4
LDR	A0	ADC Channel 0
LCD RS	Pin 7	PH4
LDR E	Pin 6	PH3
LCD D4-D7	Pins 3, 4, 5, 8	PE5, PG5, PE3, PH5
RTC SDA	Pin 20	12C SDA
RTC SCL	Pin 21	12C SCL
LED OFF (Red)	Pin 22	PA0
LED IDLE (Yellow)	Pin 23	PA1
LED ACTIVE (Green)	Pin 24	PA2
LED ERROR (Blue)	Pin 25	PA3
Room LED	Pin 13	PB7
ON Button	Pin 2	PE4 (INT4)

OFF Button	Pin 30	PC7
Reset Button	Pin 12	PB6

2.2 Component List with References

Component	Quantity	Description	References
ATMega 2560	1	Microcontroller, 54 digital pins	Elegoo Mega 2560 R3
HC-SR04	1	Ultrasonic sensor, range 2 - 400 cm	howtomechatronics.com
Photoresistor (LDR)	1	Light dependent resistor	brainy-bits.com
10kΩ Resistor	1	Voltage divider for LDR	Course kit
16x2 LCD Display	1	HD44780 compatible display	dronebotworkshop.com
10kΩ Potentiometer	1	LCD contrast control	Course kit
RTC DS1307	1	Real time clock	dronebotworkshop.com
Push Button	3	Momentary tactile switch	Course kit
LED (R, Y, G, B, W)	5	State Indicators + room light actuator	Course kit
220Ω Resistor	5	LED current limiting	Course kit
Breadboard	1	Circuit connection	Course kit
Jumper Wires	~30	Circuit connection	Course kit
USB Cable	1	Programming and power supply	Course kit

3. System Behavior

3.1 Module Design

The system is divided into four distinct software modules, each responsible for a specific function:

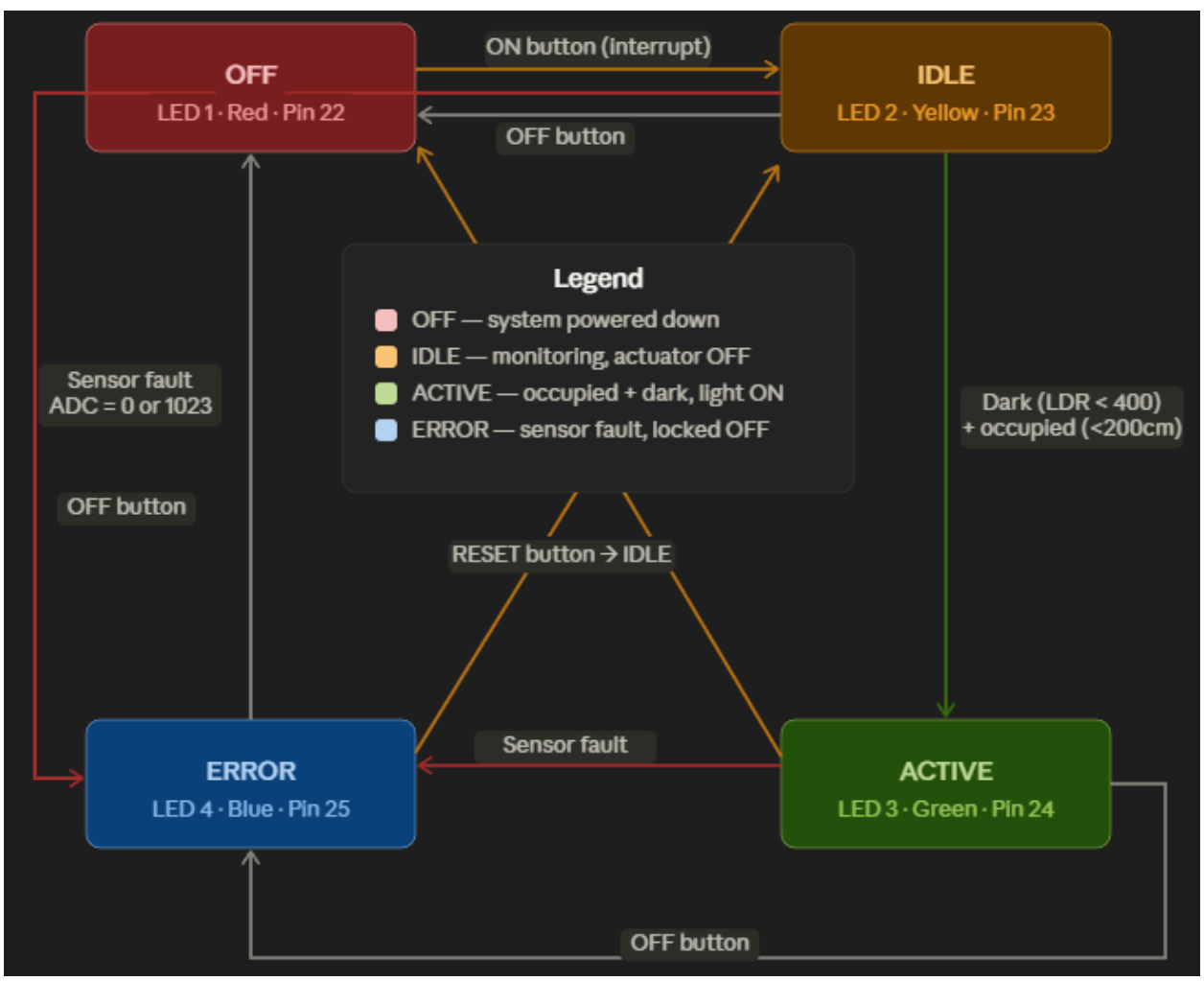
Sensing Module: This module will be in charge of the acquisition of the environmental variables. This involves obtaining ADC readings based on the analog value produced by the photoresistor by directly manipulating the ADC registers (ADCSRA and ADMUX), and measuring distance by sending pulses to the HC-SR04 ultrasonic sensor through the PH6 (TRIG) and PB4 (ECHO) pins.

Control Logic Module (FSM): This is the core module that determines transitions between states: OFF, IDLE, ACTIVE, and ERROR. It processes inputs from the sensing module and button readings, applies threshold logic (LDR < 400 = dark, distance < 200cm = occupied), manages a 3-second fault confirmation timer, and calls enterState() to transition between states. The ON button is handled via an external interrupt ISR (attachInterrupt on Pin 2).

Display Module: This module updates the 16x2 LCD display using the LiquidCrystal library and controls the 4 state indicator LEDs via direct PORTA register manipulation (PA0–PA3). It shows the current state name on row 1 and live sensor readings (distance and LDR value) on row 2, updated every 1 minute using millis().

Logging Module: This module communicates with the RTC DS1307 via 12C (Wire library) on Mega pins 20 (SDA) and 21 (SCL). It formats timestamped strings and transmits them to the Serial Monitor using direct UART register access (UCSR0, UDR0) at 9600 baud. Every state transition and sensor fault is logged with a precise timestamp.

3.2 Control Logic (Diagram)



Detailed Logic and Hardware Interaction:

Interrupt Handling (ON Button): The ON button is connected to Pin 2 (PE4) which supports external interrupt INT4 on the Mega. The interrupt is configured using attachInterrupt(digitalPinToInterrupt(2), onButtonISR, FALLING). The ISR sets a volatile flag which is checked in loop(). This enables instant response without polling.

Sensor Interaction: The system reads the photoresistor via ADC channel 0 every 60 seconds using millis()-based timing. If the ADC value is below 400 (dark) AND the ultrasonic sensor detects an object within 200cm, the system transitions from IDLE to ACTIVE.

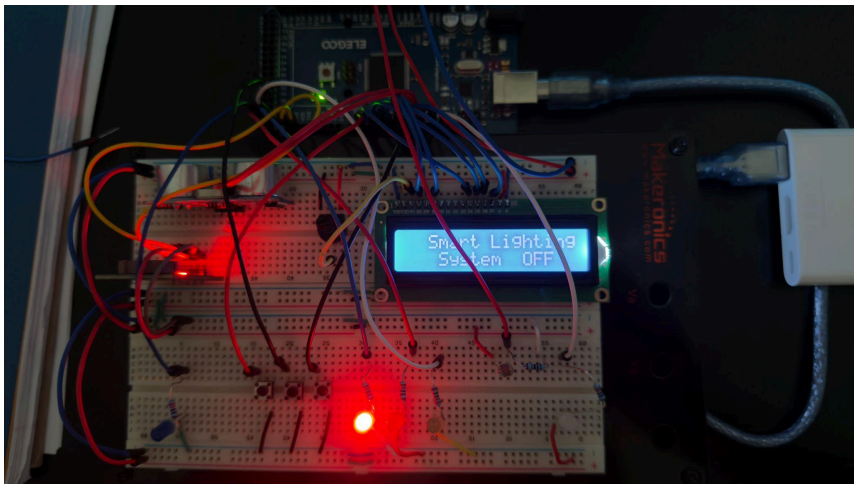
Actuator Interaction: The room LED (Pin 13, PB7) is exclusively turned on in ACTIVE state using SET_HIGH(PORTB, 7). It is immediately turned off upon any transition out of the ACTIVE state.

Error and Safety Logic: If the LDR reads 0 or 1023 (impossible values indicating sensor disconnection) for more than 3 seconds continuously, the system enters ERROR state and locks the actuator OFF. This prevents the light from staying on indefinitely due to a faulty sensor.

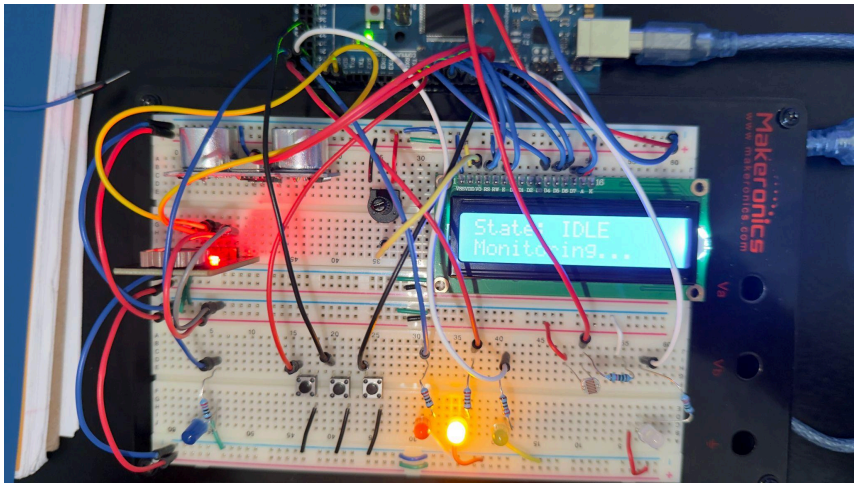
Reset Behavior: Pressing RESET while in ERROR state clears all fault flags, turns off all LEDs, transitions to IDLE state, and updates the LCD display.

Results

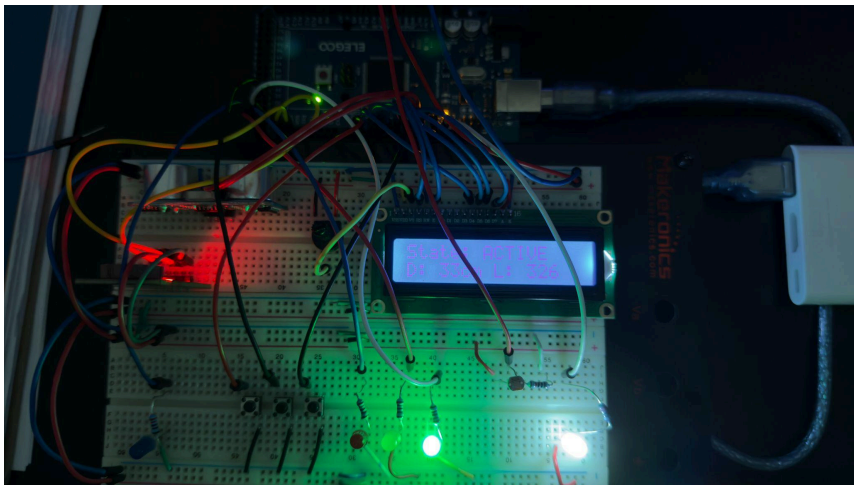
OFF State



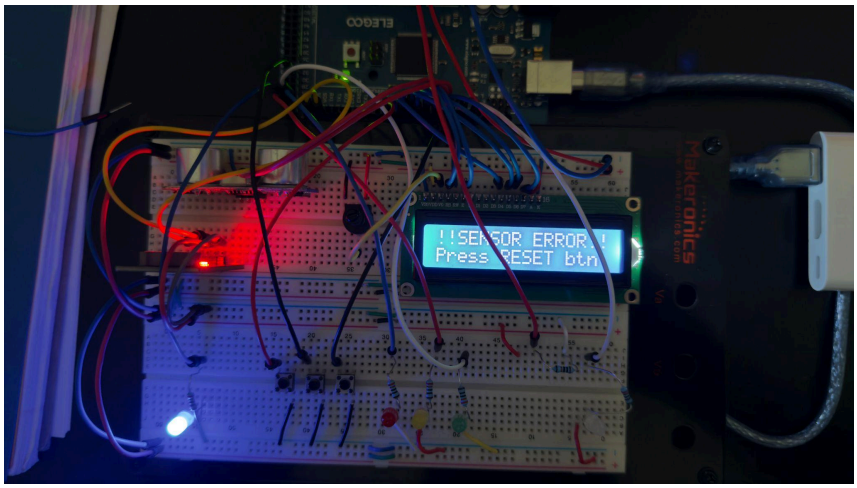
IDLE State



ACTIVE State



ERROR State



Serial Monitor

```
[26/05/01 08:13:46] System initialized
[26/05/01 08:13:50] STATE: OFF
[26/05/01 08:13:50] System initialized
[26/05/01 08:13:55] STATE: IDLE
[26/05/01 08:13:59] Dist:35.1cm LDR:72 OCC:YES DARK:YES
[26/05/01 08:13:59] STATE: ACTIVE
[26/05/01 08:13:59] Dist:35.1cm LDR:323 OCC:YES DARK:YES
[26/05/01 08:14:03] Dist:33.8cm LDR:350 OCC:YES DARK:YES
[26/05/01 08:14:07] Dist:33.8cm LDR:801 OCC:YES DARK:NO
[26/05/01 08:14:07] STATE: IDLE
[26/05/01 08:14:07] Dist:33.8cm LDR:791 OCC:YES DARK:NO
[26/05/01 08:14:11] Dist:33.8cm LDR:790 OCC:YES DARK:NO
[26/05/01 08:14:15] Dist:37.3cm LDR:1 OCC:YES DARK:YES
[26/05/01 08:14:15] Sensor fault detected
[26/05/01 08:14:15] STATE: ACTIVE
[26/05/01 08:14:15] Dist:37.3cm LDR:2 OCC:YES DARK:YES
[26/05/01 08:14:19] Dist:36.9cm LDR:2 OCC:YES DARK:YES
[26/05/01 08:14:19] Fault confirmed - entering ERROR
[26/05/01 08:14:20] STATE: ERROR
[26/05/01 08:14:25] STATE: IDLE
[26/05/01 08:14:28] STATE: OFF
```

4. Environmental and Energy Impact

Energy Efficiency: The system's threshold-based logic ensures the room LED is only powered when both occupancy and darkness conditions are simultaneously met. The FSM architecture is inherently energy-efficient — the actuator remains OFF in three out of four states (OFF, IDLE, ERROR). By eliminating unnecessary lighting in empty or naturally-lit rooms, the system can reduce residential lighting energy consumption significantly over time.

Design Safety: The system operates entirely on 5V DC supplied via USB, eliminating any risk of high-voltage exposure. All LEDs are protected with 220Ω current-limiting resistors to prevent overcurrent damage. The ERROR state provides a critical safety mechanism — when sensors report impossible values the system locks the actuator OFF and requires manual RESET, preventing the light from remaining on indefinitely due to a hardware fault.

Affordability: The entire project is built using components from the CPE 301 course kit, making it extremely cost-effective. The total component cost is estimated at under $75. The use of an Arduino Mega instead of a custom PCB significantly reduces development cost while maintaining full functionality.

Sustainability: The system's modular design means individual components can be replaced without discarding the entire unit, reducing electronic waste. The breadboard-based construction allows for easy repair and component substitution. The use of an LED to simulate room lighting instead of actual high-voltage bulbs makes the system safe and reusable across multiple test environments.

Accessibility: The 16x2 LCD display provides real-time feedback to the user about the system's current state and sensor readings, making the system transparent and easy to understand. The three clearly labeled push buttons (ON, OFF, RESET) provide a simple and intuitive user interface that requires no technical knowledge to operate.

5. Conclusion

The Smart Energy-Saving Room Lighting System was successfully implemented and demonstrated all four FSM states (OFF, IDLE, ACTIVE, ERROR) with correct transitions. The ultrasonic sensor accurately detected occupancy within 200cm, the photoresistor correctly measured ambient light levels, the LCD display updated in real time, the RTC logged all state transitions with timestamps, and all three buttons functioned as designed. The system successfully turned on the room LED only when the room was both occupied and dark, achieving the core energy-saving objective.

Limitations

- The ultrasonic sensor has a narrow detection cone which may miss occupants not directly in front of it.

- The 60-second polling interval means the system can take up to 1 minute to respond to environmental changes.
- The system uses an LED to simulate room light rather than controlling actual high-voltage lighting.
- The LDR is sensitive to direct light sources which can cause incorrect readings.